Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	No live weather API integration	Users must manually enter all weather values	High
2	Not deployed to the cloud	Only accessible via local Flask server	Medium
3	Model depends on historical dataset quality	Accuracy limited by training data coverage	Medium

Scalability Plan

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local Flask dev server	Deploy with Gunicorn + Nginx or a managed cloud platform
2	Data Storage	Static Excel file	Migrate to a database (e.g., PostgreSQL) for larger datasets
3	Performance	Single-threaded inference	Add caching and asynchronous request handling
4	Security	No authentication (open demo)	Add basic auth/rate-limiting if made public

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Live weather API integration (e.g., OpenWeatherMap)	Q3 2026	Removes manual data entry, more accurate real-time predictions
Phase 3	Cloud deployment (Render/Heroku/AWS)	Q4 2026	Public accessibility for wider use
Phase 4	GIS-based flood mapping & mobile app	2027	Visual, location-based risk representation for broader adoption